Practical - 11

Practical Title: Student Management CRUD with Search and Filter

Problem definition: Create a student management module to add, view, edit, delete, search and filter student records. All database operations must use MYSQLi statements and clear success/failure messages.

Theory:

1. Are SELECT, INSERT, UPDATE, and DELETE operations working?

   Answer: Yes, all CRUD operations are working correctly. SELECT displays student records, INSERT adds new students, UPDATE modifies student details, and DELETE removes student records.

2. Is search by name/email/course implemented?

   Answer: Yes, search by student name, email, and course is implemented. The course filter also displays students from the selected course.

3. Are delete operations confirmed before execution?

   Answer: Yes, a confirmation message appears before deleting a student record. The record is deleted only after clicking OK.

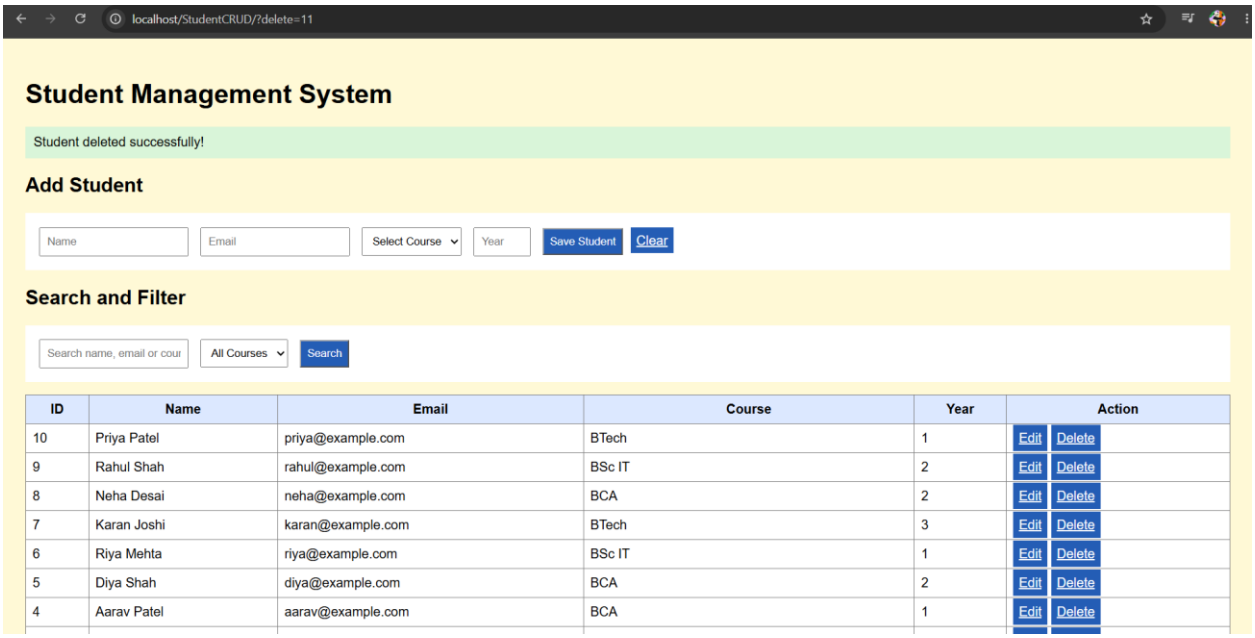
A. Implementation/ Output/Result **(Keep only one option which is suitable to practical)**

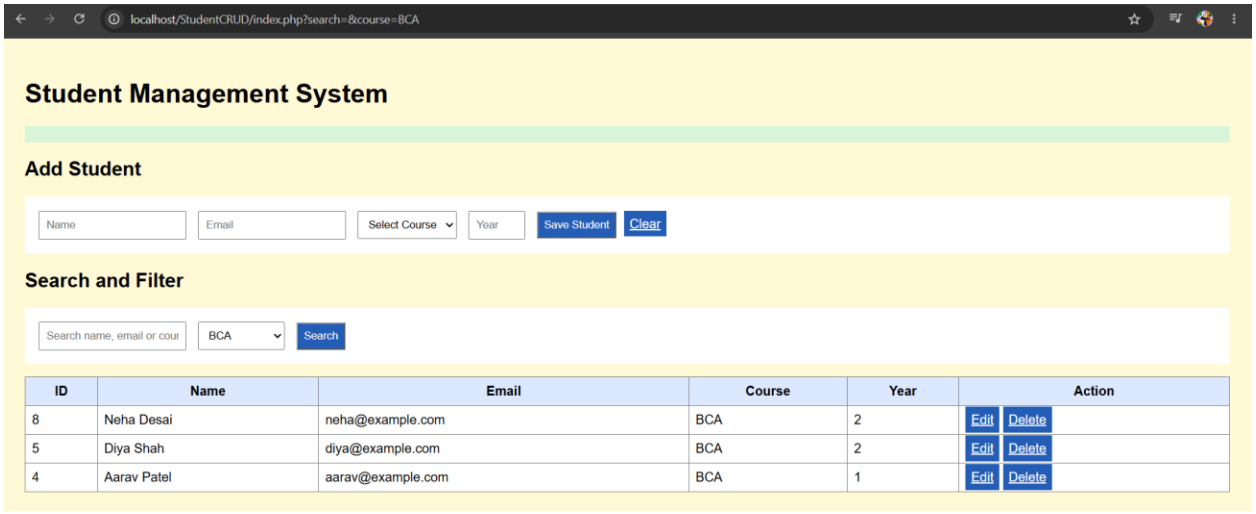
PHP Functions and MySQLi Methods Used

Function / Method	Purpose
new mysqli()	Connects PHP to MySQL.
prepare()	Prepares an SQL statement.
bind_param()	Binds values to query parameters.
execute()	Executes the SQL statement.
get_result()	Retrieves query results.
fetch_assoc()	Fetches a row as an associative array.
trim()	Removes extra whitespace from input.
htmlspecialchars()	Helps safely display user-provided text.
confirm()	Asks for confirmation before deletion.
isset()	Checks whether a variable or form value exists.

Code	Purpose
$conn = new mysqli(...)	Connects PHP to MySQL
INSERT INTO	Adds a new student
SELECT	Retrieves student records
UPDATE	Modifies student information
DELETE FROM	Removes a student
$conn->prepare()	Prepares an SQL statement safely
bind_param()	Binds values to the query parameters
LIKE	Searches for matching text
LIMIT and OFFSET	Display records page by page
confirm()	Asks the user before deletion
htmlspecialchars()	Helps prevent HTML/script injection when displaying data
filter_var()	Validates the email format

B. Evidence of Work (Screenshots)





C. GitHub Repository Link:

https://github.com/diyatolia1212/StudentHub/tree/02888d8eedb05527e7f8fc5caea244c41e85e110/StudentCRUD

D. Challenges Faced:

1. Connecting PHP with the MySQL database using MySQLi.
2. Performing INSERT, SELECT, UPDATE, and DELETE operations correctly.
3. Fixing the mismatch between the `id` and `student_id` column names.
4. Implementing search by name, email, and course.
5. Displaying proper success and error messages.
6. Adding a confirmation message before deleting student records.

Learning Outcomes:

1. Learned how to connect PHP with a MySQL database using MySQLi.
2. Understood how to perform CRUD operations on student records.
3. Learned to search students by name, email, and course.
4. Understood how to filter records based on the selected course.
5. Learned to use prepared statements for safer database queries.

Faculty Remarks: (Keep this blank for faculty)